**4.1**

**(a)** $T_1 = B'C$, $T_2 = A'B$, $T_3 = A + T_1 = A + B'C$,

$T_4 = D \oplus T_2 = D \oplus (A'B) = A'BD' + D(A + B') = A'BD' + AD + B'D$

$F_1 = T_3 + T_4 = A + B'C + A'BD' + AD + B'D$

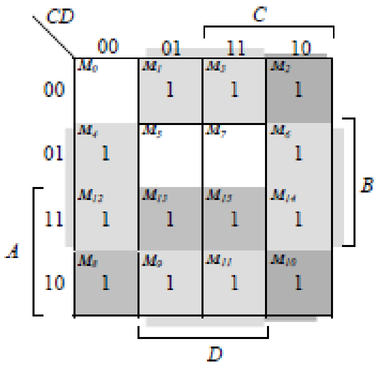
With $A + AD = A$ and $A + A'BD' = A + BD'$:

$F_1 = A + B'C + BD' + B'D$

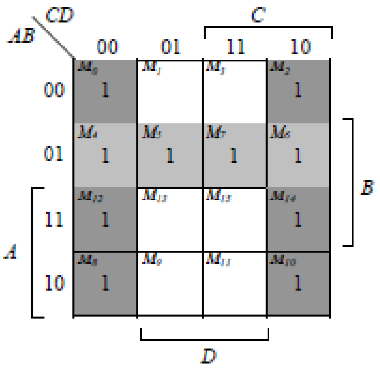
Alternative cover: $F_1 = A + CD' + BD' + B'D$

$F_2 = T_2 + D' = A'B + D'$

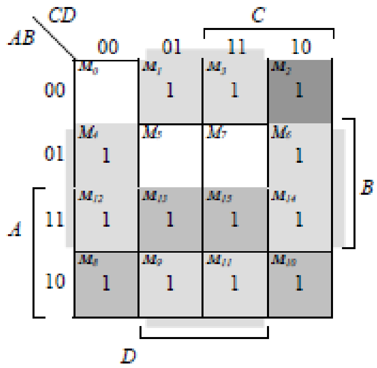
| ABCD | $T_1$ | $T_2$ | $T_3$ | $T_4$ | $F_1$ | $F_2$ |
|---|---|---|---|---|---|---|
| 0000 | 0 | 0 | 0 | 0 | 0 | 1 |
| 0001 | 0 | 0 | 0 | 1 | 1 | 0 |
| 0010 | 1 | 0 | 1 | 0 | 1 | 1 |
| 0011 | 1 | 0 | 1 | 1 | 1 | 0 |
| 0100 | 0 | 1 | 0 | 1 | 1 | 1 |
| 0101 | 0 | 1 | 0 | 0 | 0 | 1 |
| 0110 | 0 | 1 | 0 | 1 | 1 | 1 |
| 0111 | 0 | 1 | 0 | 0 | 0 | 1 |
| 1000 | 0 | 0 | 1 | 0 | 1 | 1 |
| 1001 | 0 | 0 | 1 | 1 | 1 | 0 |
| 1010 | 1 | 0 | 1 | 0 | 1 | 1 |
| 1011 | 1 | 0 | 1 | 1 | 1 | 0 |
| 1100 | 0 | 0 | 1 | 0 | 1 | 1 |
| 1101 | 0 | 0 | 1 | 1 | 1 | 0 |
| 1110 | 0 | 0 | 1 | 0 | 1 | 1 |
| 1111 | 0 | 0 | 1 | 1 | 1 | 0 |



$F_1 = A + B'C + B'D + BD'$

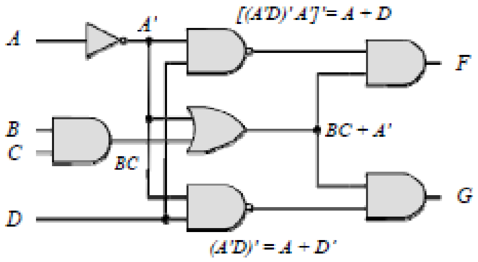


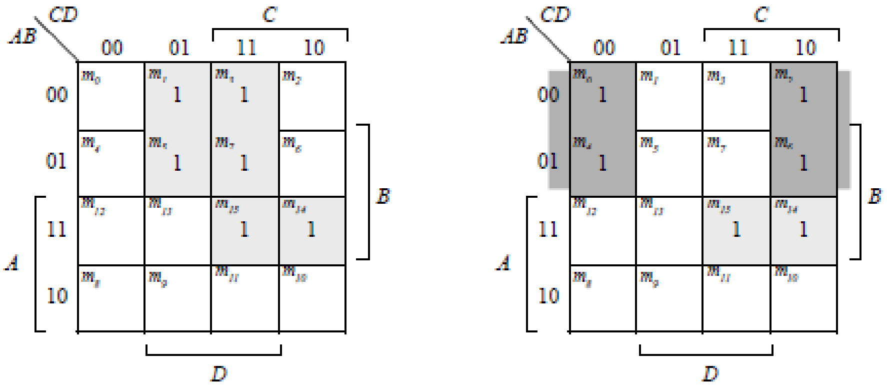
$F_2 = A'B + D'$



$F_1 = A + CD' + B'D + BD'$

**4.2**



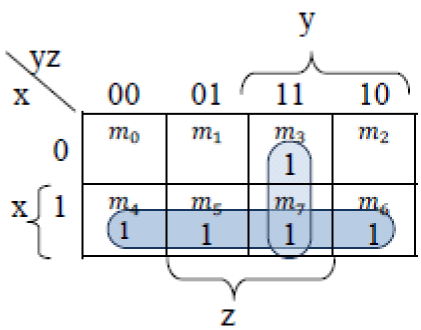
$F = A'D + ABC + BCD = A'D + ABC$ $\qquad$ $G = A'D' + ABC + BCD' = A'D' + ABC$

**4.3** (a) $Y_i = (A_iS' + B_iS)E'$ *for* $i = 0, 1, 2, 3$

(b) 1024 rows and 14 columns

**4.4** (a) Circuit where output is 1 when the binary value of the inputs is more than 2. The output is 0 otherwise.

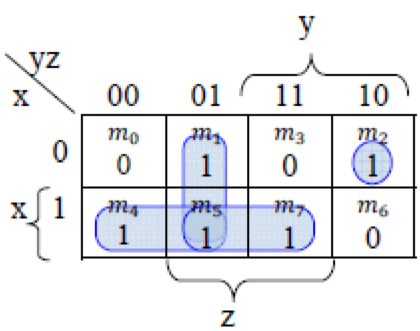
| $x$ | $y$ | $z$ | $F$ |
|---|---|---|---|
| 0 | 0 | 0 | 0 |
| 0 | 0 | 1 | 0 |
| 0 | 1 | 0 | 0 |
| 0 | 1 | 1 | 1 |
| 1 | 0 | 0 | 1 |
| 1 | 0 | 1 | 1 |
| 1 | 1 | 0 | 1 |
| 1 | 1 | 1 | 1 |



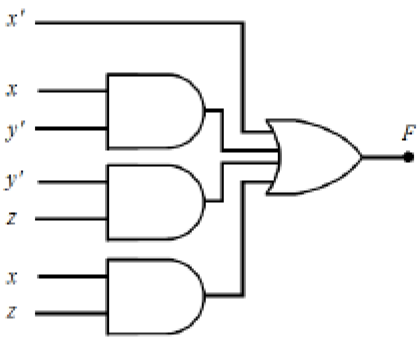

$F = x + yz$


**(b)** The output is 1 when the binary value of the inputs is not divisible by 3.

| $x$ | $y$ | $z$ | $F$ |
|---|---|---|---|
| 0 | 0 | 0 | 0 |
| 0 | 0 | 1 | 1 |
| 0 | 1 | 0 | 1 |
| 0 | 1 | 1 | 0 |
| 1 | 0 | 0 | 1 |
| 1 | 0 | 1 | 1 |
| 1 | 1 | 0 | 0 |
| 1 | 1 | 1 | 1 |

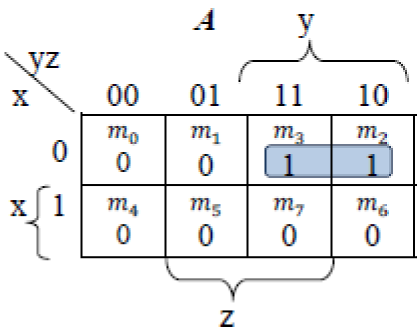



$F = x' + xy' + xz + y'z$

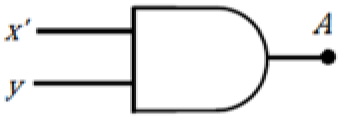


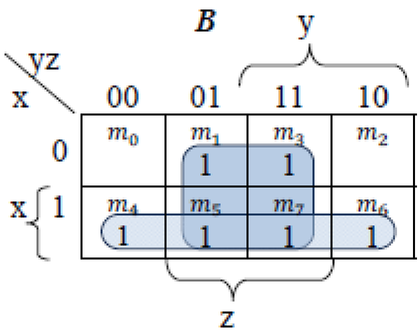
**4.5**

| $x$ | $y$ | $z$ | $A$ | $B$ | $C$ |
|---|---|---|---|---|---|
| 0 | 0 | 0 | 0 | 0 | 0 |
| 0 | 0 | 1 | 0 | 1 | 0 |
| 0 | 1 | 0 | 1 | 0 | 0 |
| 0 | 1 | 1 | 1 | 1 | 0 |
| 1 | 0 | 0 | 0 | 1 | 0 |
| 1 | 0 | 1 | 0 | 1 | 0 |
| 1 | 1 | 0 | 0 | 1 | 1 |
| 1 | 1 | 1 | 0 | 1 | 1 |

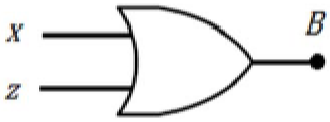


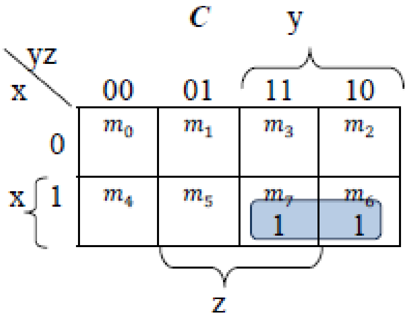

$A = x'y$



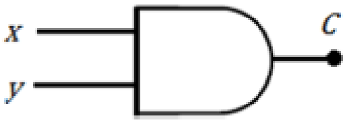




$B = x + z$



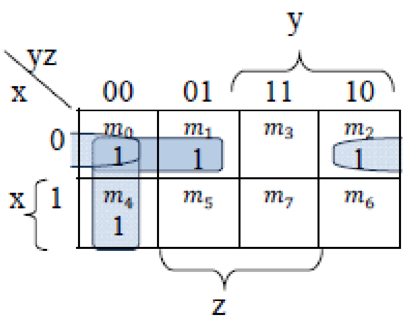



$C = xy$

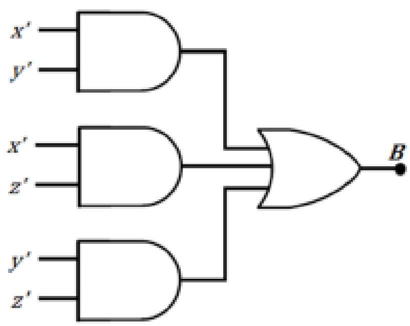



**4.6** **(a)**

| x | y | z | F |
|---|---|---|---|
| 0 | 0 | 0 | 1 |
| 0 | 0 | 1 | 1 |
| 0 | 1 | 0 | 1 |
| 0 | 1 | 1 | 0 |
| 1 | 0 | 0 | 1 |
| 1 | 0 | 1 | 0 |
| 1 | 1 | 0 | 0 |
| 1 | 1 | 1 | 0 |

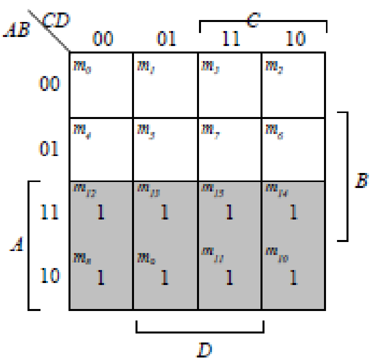



$Y = x'y' + x'z' + y'z'$

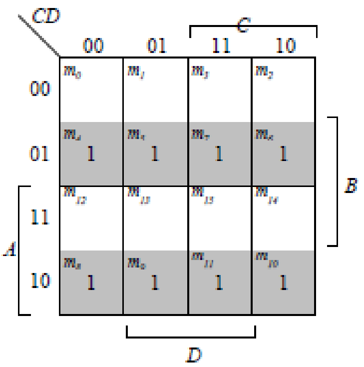


**4.7** **(a)**

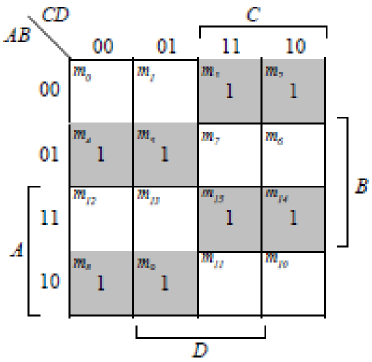
| ABCD | wxyz |
|---|---|
| 0000 | 0000 |
| 0001 | 0001 |
| 0011 | 0010 |
| 0010 | 0011 |
| 0110 | 0100 |
| 0111 | 0101 |
| 0101 | 0110 |
| 0100 | 0111 |
| 1100 | 1000 |
| 1101 | 1001 |
| 1111 | 1010 |
| 1110 | 1011 |
| 1010 | 1100 |
| 1011 | 1101 |
| 1001 | 1110 |
| 1000 | 1111 |



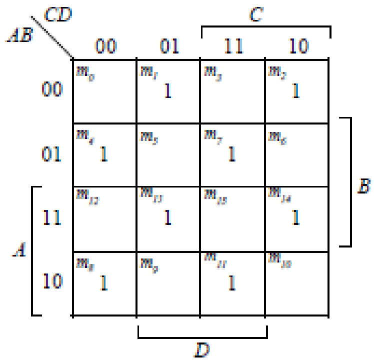

$w = A$



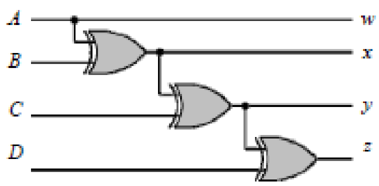

$x = AB' + A'B = A \oplus B$




$y = A'B'C\ A'BC' + ABC + AB'C'$
$= A'(A \oplus B) + A(B \oplus C)'$
$= A \oplus B \oplus C$
$= X \oplus C$



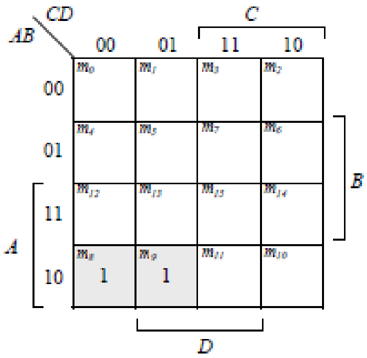

$z = A \oplus B \oplus C \oplus D$
$= y \oplus D$



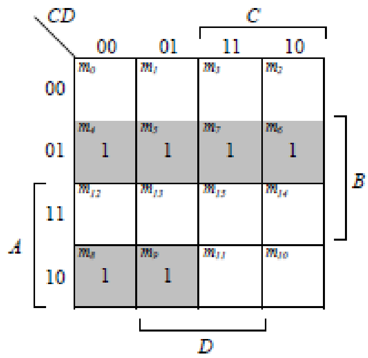
**4.8** (a) The 8-4-2-1 code (Table 1.5) and the BCD code (Table 1.4) are identical for digits 0 – 9.

**(b)**

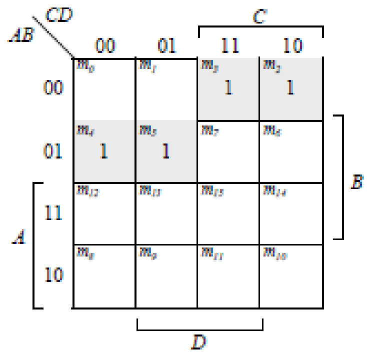
| *8421* *ABCD* | *Gray* *wxyz* |
|---|---|
| 0000 | 0000 |
| 0001 | 0001 |
| 0010 | 0011 |
| 0011 | 0010 |
| 0100 | 0110 |
| 0101 | 0111 |
| 0110 | 0101 |
| 0111 | 0100 |
| 1000 | 1100 |
| 1001 | 1101 |



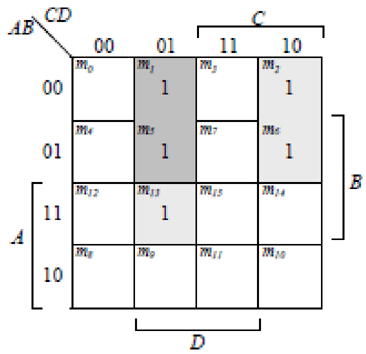
$w = AB'C'$



$x = AB'C' + A'B$



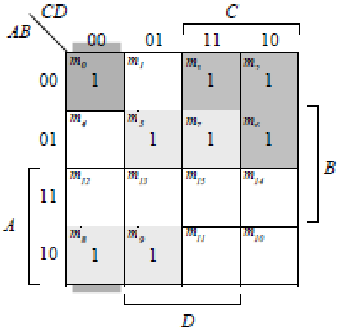
$y = A'BD' + A'B'D$



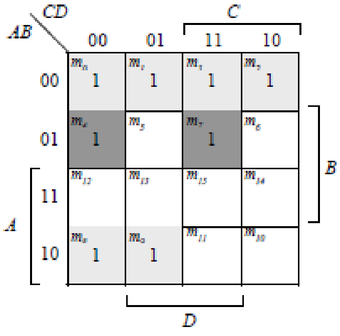
$z = A'C'D + BC'D + A'CD'$

**4.9**

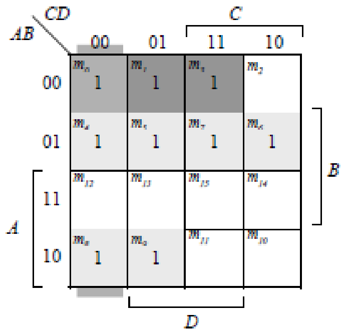
| ABCD | a | b | c | d | e | f | g |
|---|---|---|---|---|---|---|---|
| 0000 | 1 | 1 | 1 | 1 | 1 | 1 | 0 |
| 0001 | 0 | 1 | 1 | 0 | 0 | 0 | 0 |
| 0010 | 1 | 1 | 0 | 1 | 1 | 0 | 1 |
| 0011 | 1 | 1 | 1 | 1 | 0 | 0 | 1 |
| 0100 | 0 | 1 | 1 | 0 | 0 | 1 | 1 |
| 0101 | 1 | 0 | 1 | 1 | 0 | 1 | 1 |
| 0110 | 1 | 0 | 1 | 1 | 1 | 1 | 1 |
| 0111 | 1 | 1 | 1 | 0 | 0 | 0 | 0 |
| 1000 | 1 | 1 | 1 | 1 | 1 | 1 | 1 |
| 1001 | 1 | 1 | 1 | 1 | 0 | 1 | 1 |



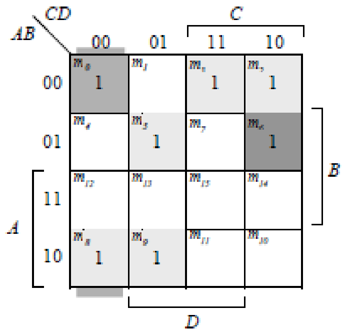
$a = A'C + A'BD + B'C'D' + AB'C'$



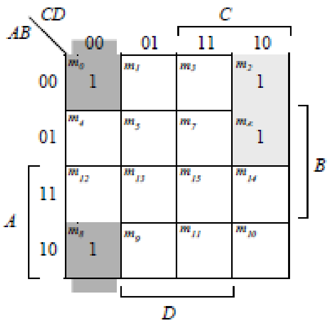
$b = A'B' + A'C'D' + A'CD + AB'C'$



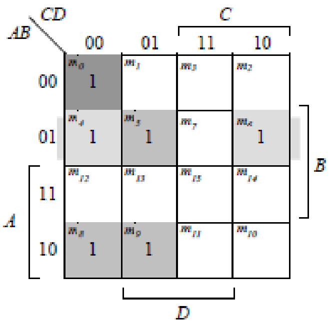
$c = A'B + A'D + B'C'D' + AB'C'$



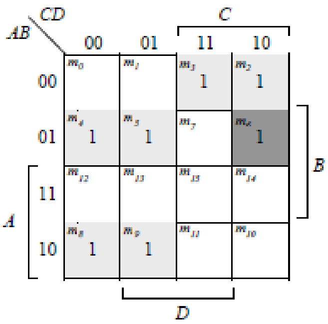
$d = A'CD' + A'B'C + B'C'D' + AB'C' + A'BC'D$



$e = A'CD' + B'C'D'$



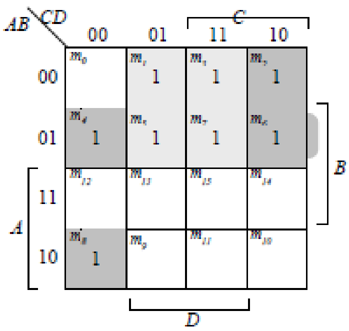
$f = A'BC' + A'C'D' + A'BD + AB'C'$



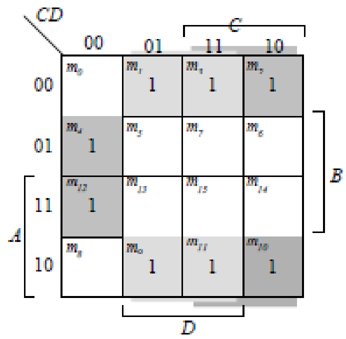
$g = A'CD' + A'B'C + A'BC' + AB'C'$

**4.10**

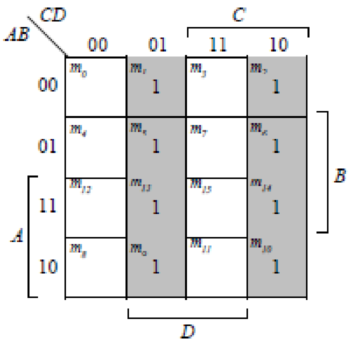
| ABCD | wxyz |
|---|---|
| 0000 | 0000 |
| 0001 | 1111 |
| 0010 | 1110 |
| 0011 | 1101 |
| 0100 | 1100 |
| 0101 | 1011 |
| 0110 | 1001 |
| 0111 | 1000 |
| 1000 | 1000 |
| 1001 | 0111 |
| 1010 | 0110 |
| 1011 | 0101 |
| 1100 | 0100 |
| 1101 | 0011 |
| 1110 | 0010 |
| 1111 | 0001 |



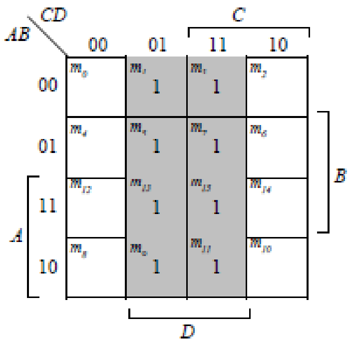
$w = A'(B + C + D) + AB'C'D'$
$= A \oplus (B + C + D)$



$x = B'(C + D) + CB'D'$
$= B \oplus (C + D)$



$y = CD' + C'D = C \oplus D$

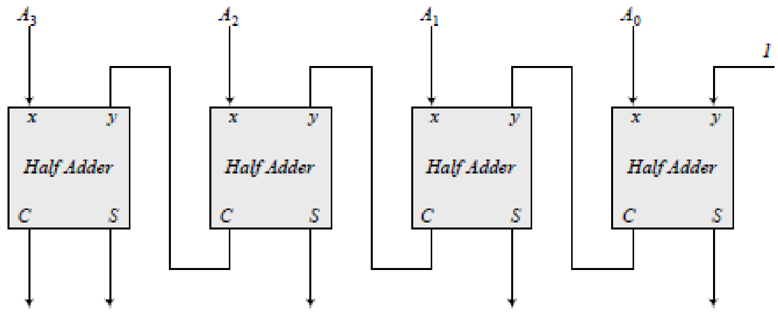


$z = D$

For a 5-bit 2's complementer with input E and output v:

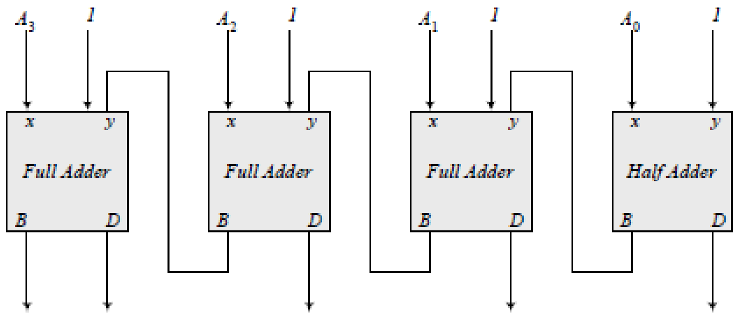
$v = E \oplus (A + B + C + D)$

**4.11** **(a)**




Note: 5-bit output

**(b)**




Note: To decrement the 4-bit number, add -1 to the number. In 2's complement format ( add $F_h$) to the number. An attempt to decrement 0 will assert the borrow bit. For waveforms, see solution to Problem 4.52.

**4.12**

**(a)**

| x y | B D |
|---|---|
| 0 0 | 0 0 |
| 0 1 | 1 1 |
| 1 0 | 0 1 |
| 1 1 | 0 0 |

$D = x'y + xy'$
$B = x'y$

**(b)**

| x y $B_{in}$ | B D |
|---|---|
| 0 0 0 | 0 0 |
| 0 0 1 | 1 1 |
| 0 1 0 | 1 1 |
| 0 1 1 | 1 0 |
| 1 0 0 | 0 1 |
| 1 0 1 | 0 0 |
| 1 1 0 | 0 0 |
| 1 1 1 | 1 1 |

$Diff = x \oplus y \oplus z$
$B_{out} = x'y + x'z + yz$

**4.13**

| | Sum | $C$ | $V$ |
|---|---|---|---|
| **(a)** | 1101 | 0 | 1 |
| **(b)** | 0001 | 1 | 1 |
| **(c)** | 0100 | 1 | 0 |
| **(d)** | 1011 | 0 | 1 |
| **(e)** | 1111 | 0 | 0 |

**4.14** xor AND OR XOR

10 + 5 + 5 + 10 = 30 ns

**4.15** $C_4 = G_3 + P_3G_2 + P_3P_2G_1 + P_3P_2P_1G_0 + P_3P_2P_1P_0C_0$

**4.16** **(a)**

$(C'G'_i + p'_i)' = A_iB_i + (A_i + B_i)C_i = A_iB_i + A_iC_i + B_iC_i = C_{i+1}$

$(P_iG'_i) \oplus C_i = (A'_iB_i + A_iB'_i) \oplus C_i = A_i \oplus B_i \oplus C_i = S_i$

**(b)**

Output of NOR gate = $(A_0 + B_0)' = P'_0$
Output of NAND gate = $(A_0B_0)' = G'_0$
$S_1 = (P_0G'_0) \oplus C_0$
$C_1 = (C'_0G'_0 + P'_0)'$ as defined in part (a)

**4.17** **(a)**

$(C'_iG'_i + P'_i)' = A_iB_i + (A_i + B_i)C_i = A_iB_i + A_iC_i + B_iC_i = C_{i+1}$

$(P_iG'_i)\oplus C_i = (A'_iB_i + A_iB'_i)\oplus C_i = A_i\oplus B_i\oplus C_i = S_i$

**(b)**

Output of NOR gate = $(A_0 + B_0)' = P'_0$

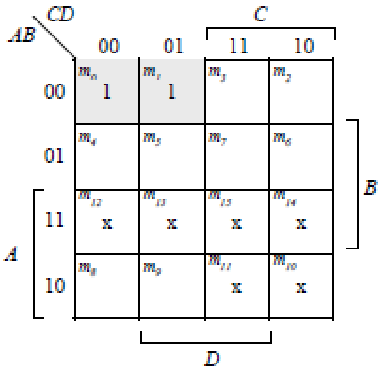
Output of NAND gate = $(A_0B_0)' = G'_0$

$S_0 = (P_0G'_0)\oplus C_0$
$C_1 = (C'_0G'_0 + P'_0)'$ as defined in part (a)

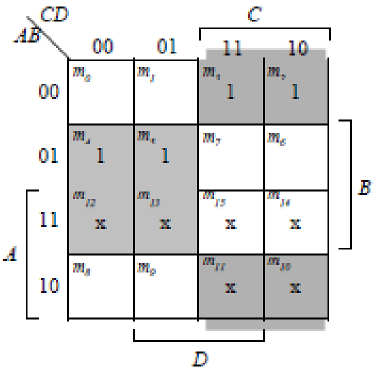
**4.18**

| Inputs *ABCD* | Outputs *wxyz* |
|---|---|
| 0000 | 1001 |
| 0001 | 1000 |
| 0010 | 0111 |
| 0011 | 0110 |
| 0100 | 0101 |
| 0101 | 0100 |
| 0110 | 0011 |
| 0111 | 0010 |
| 1000 | 0001 |
| 1001 | 0000 |

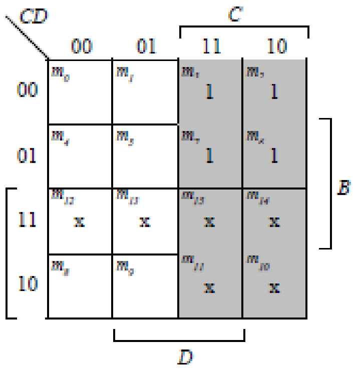
$d(A, b, c, d) = \Sigma(10, 11, 12, 13, 14, 15)$



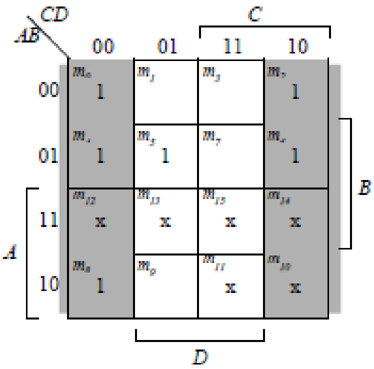

$w = A'B'C'$




$x = BC' + B'C = B \oplus C$

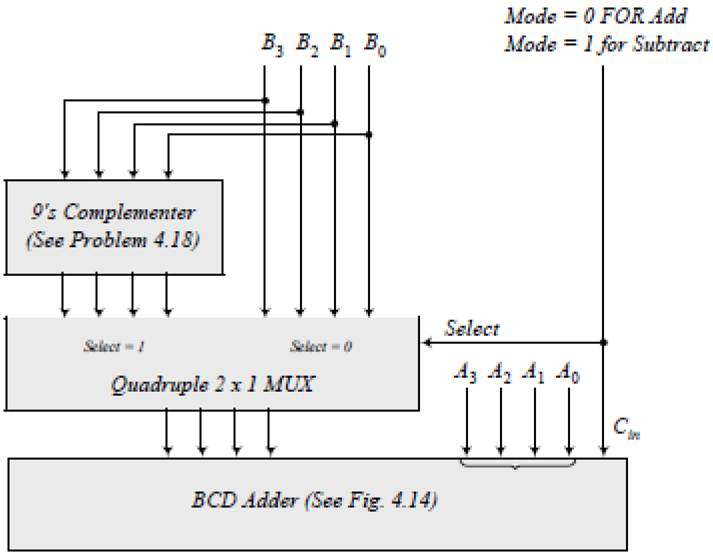



$y = C$



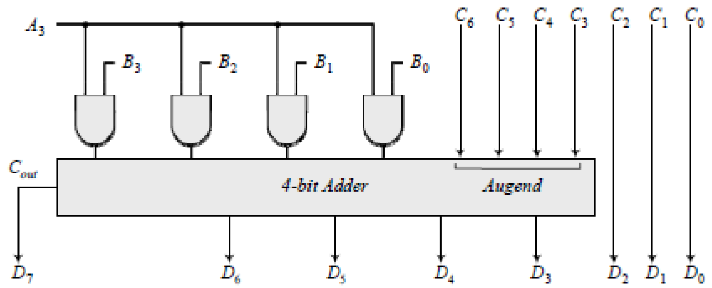

$z = D'$

**4.19**



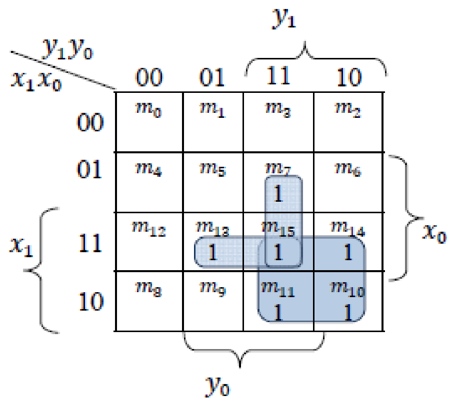
Mode = 0 FOR Add
Mode = 1 for Subtract
$B_3$ $B_2$ $B_1$ $B_0$
9's Complementer
(See Problem 4.18)
Select = 1
Select = 0
Select
Quadruple 2 x 1 MUX
$A_3$ $A_2$ $A_1$ $A_0$
$C_{in}$
BCD Adder (See Fig. 4.14)

**4.20** Combine the following circuit with the 4-bit binary multiplier circuit of Fig. 4.16.

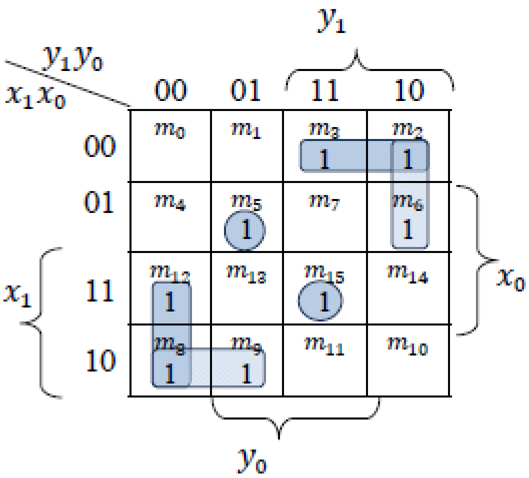



**4.21**

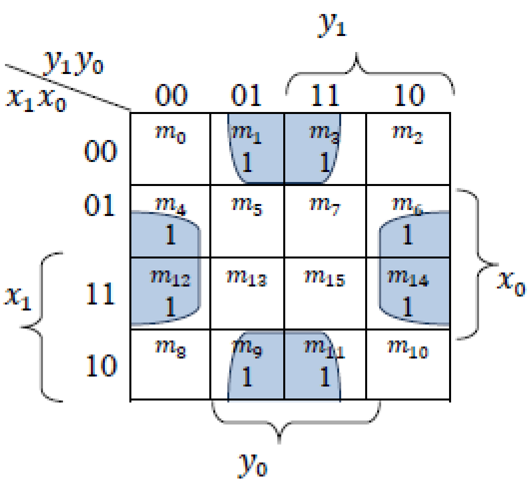
| $x_1$ | $x_0$ | $y_1$ | $y_0$ | $S_2$ | $S_1$ | $S_0$ |
|---|---|---|---|---|---|---|
| 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 0 | 0 | 0 | 1 | 0 | 0 | 1 |
| 0 | 0 | 1 | 0 | 0 | 1 | 0 |
| 0 | 0 | 1 | 1 | 0 | 1 | 1 |
| 0 | 1 | 0 | 0 | 0 | 0 | 1 |
| 0 | 1 | 0 | 1 | 0 | 1 | 0 |
| 0 | 1 | 1 | 0 | 0 | 1 | 1 |
| 0 | 1 | 1 | 1 | 1 | 0 | 0 |
| 1 | 0 | 0 | 0 | 0 | 1 | 0 |
| 1 | 0 | 0 | 1 | 0 | 1 | 1 |
| 1 | 0 | 1 | 0 | 1 | 0 | 0 |
| 1 | 0 | 1 | 1 | 1 | 0 | 1 |
| 1 | 1 | 0 | 0 | 0 | 1 | 1 |
| 1 | 1 | 0 | 1 | 1 | 0 | 0 |
| 1 | 1 | 1 | 0 | 1 | 0 | 1 |
| 1 | 1 | 1 | 1 | 1 | 1 | 0 |



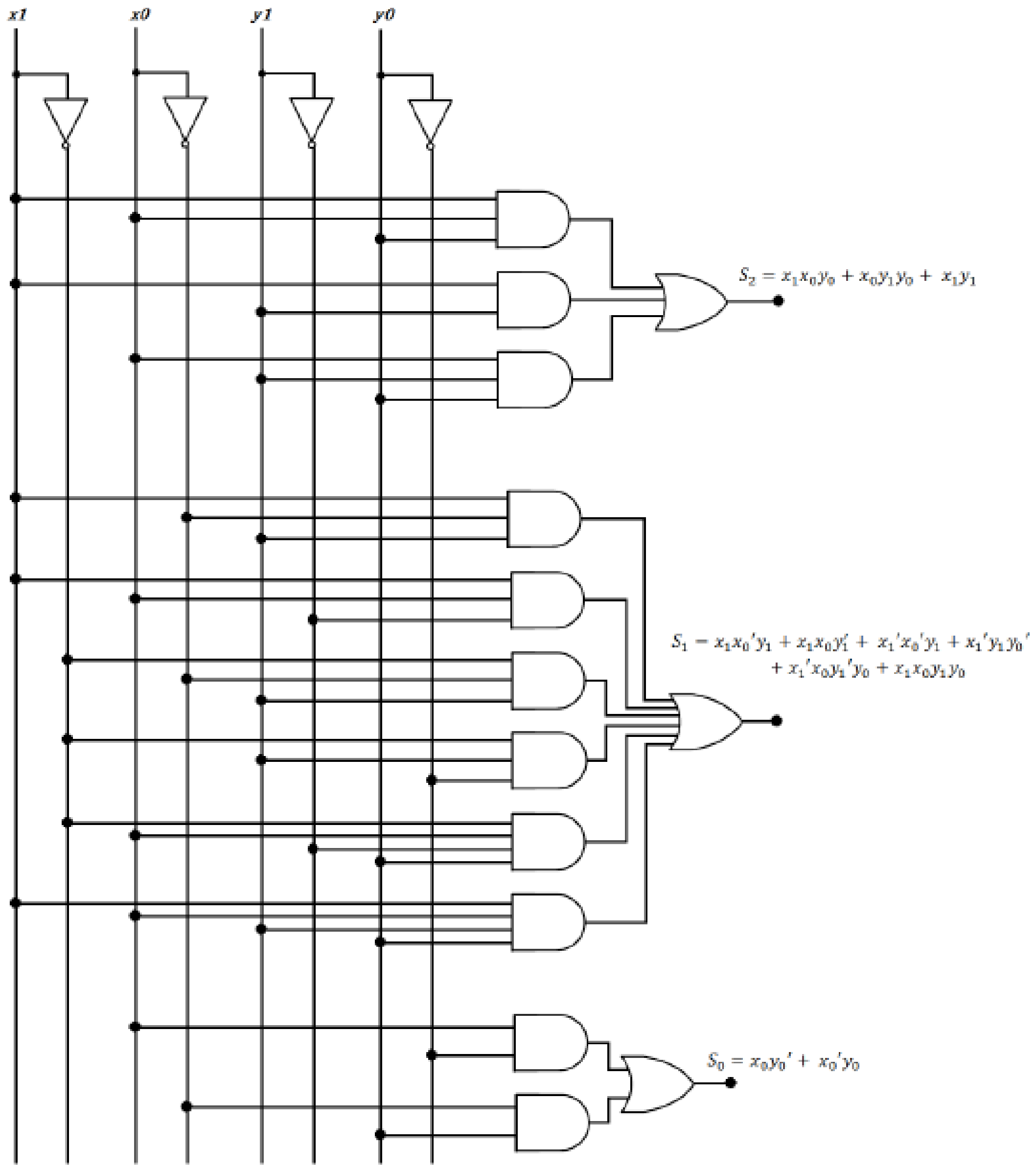

$$S_2 = x_1x_0y_0 + x_0y_1y_0 + x_1y_1$$



$$S_1 = x_1x_0'y_1' + x_1x_0'y_1' + x_1'x_0'y_1 + x_1'y_1y_0' + x_1'x_0y_1'y_0 + x_1x_0y_1y_0$$



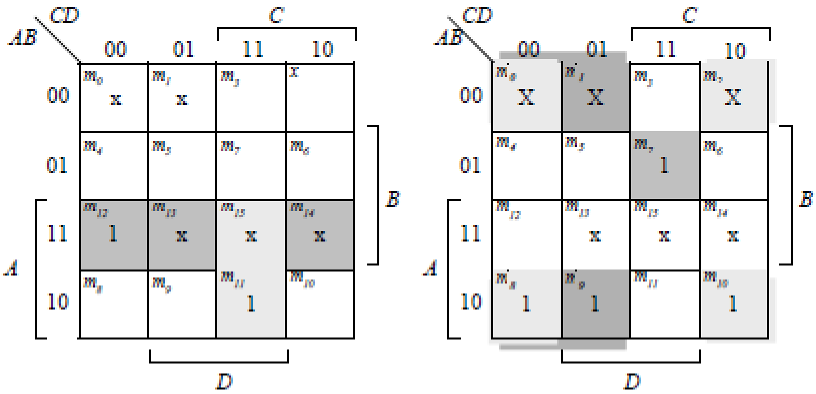
$$S_0 = x_0y_0' + x_0'y_0$$


x1
x0
y1
y0
$S_2 = x_1x_0y_0 + x_0y_1y_0 + x_1y_1$
$S_1 = x_1x_0'y_1 + x_1x_0y_1' + x_1'x_0'y_1 + x_1'y_1y_0'$
$+ x_1'x_0y_1'y_0 + x_1x_0y_1y_0$
$S_0 = x_0y_0' + x_0'y_0$

$$x = (A_0 \oplus B_0)'(A_1 \oplus B_1)'(A_2 \oplus B_2)'(A_3 \oplus B_3)'$$

**4.22**

| XS-3 *ABCD* | Binary *wxyz* |
|---|---|
| 0011 | 0000 |
| 0100 | 0001 |
| 0101 | 0010 |
| 0110 | 0011 |
| 0111 | 0100 |
| 1000 | 0101 |
| 1001 | 0110 |
| 1010 | 0111 |
| 1011 | 1000 |
| 1100 | 1001 |



$w = AB + ACD$

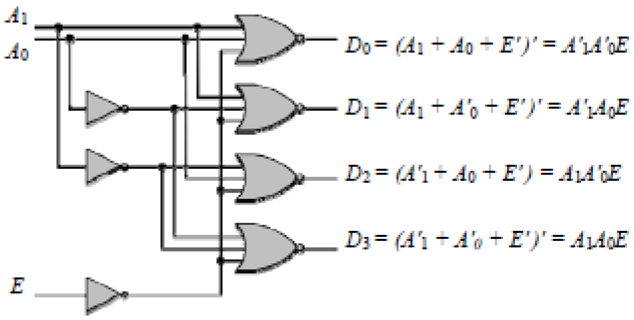
$x = B'C' + B'D' + BCD$

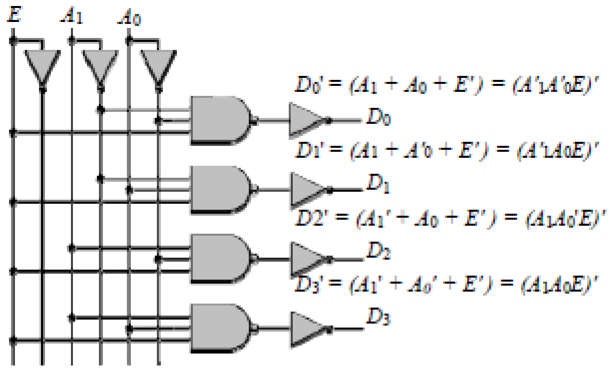
$y = C'D + CD'$

$z = D'$

**4.23**

$D0 = A1'A0' = (A1 + A0)'$ (NOR) $\quad D0' = (A1'A0')'$ (NAND)
$D1 = A1'A0 = (A1 + A0')'$ (NOR) $\quad D1' = (A1'A0)'$ (NAND)
$D2 = A1A0' = (A1' + A0)'$ (NOR) $\quad D2' = (A1A0')'$ (NAND)
$D3 = A1A0 = (A1' + A0)'$ (NOR) $\quad D0' = (A1A0)'$ (NAND)



## 4.24

Inputs: $A, B, C, D$

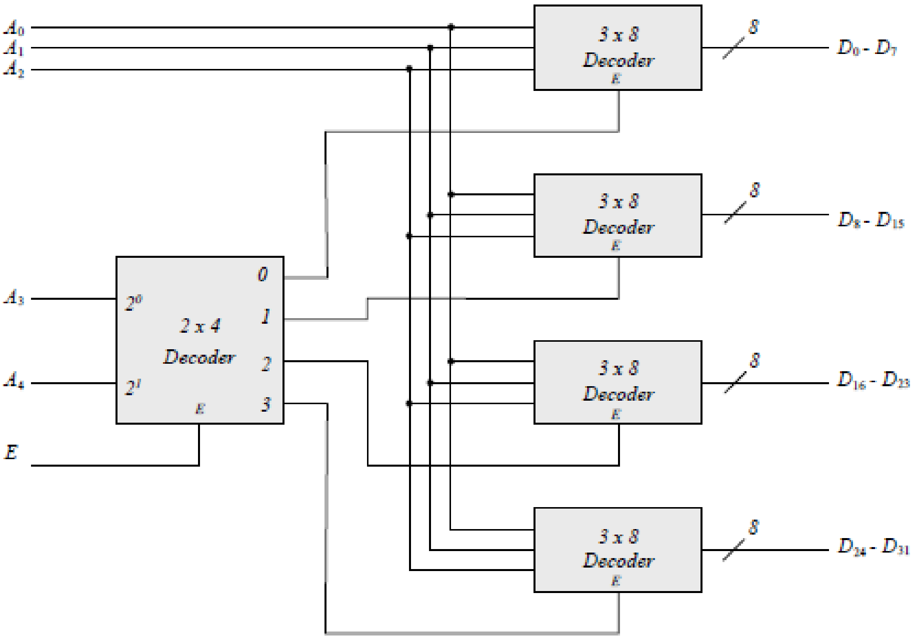
$D_0 = A'B'C'D'$
$D_1 = A'B'C'D$
$D_2 = B'CD'$
$D_3 = B'CD$
$D_4 = BC'D'$

Outputs: $D_0, D_1, \ldots D_9$

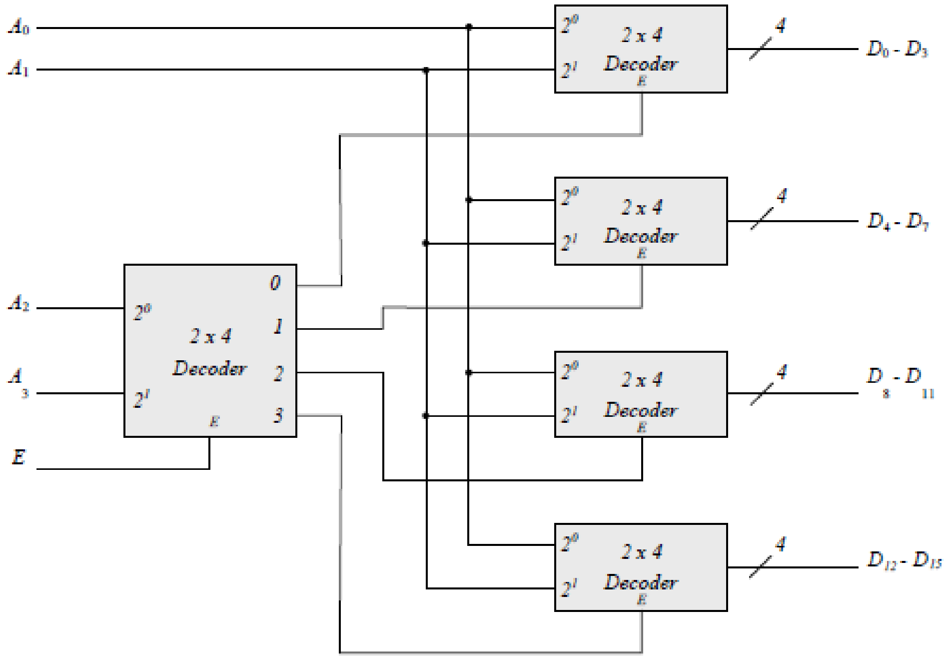
$D_5 = BC'D$
$D_6 = BCD'$
$D_7 = BCD$
$D_8 = AD'$
$D_9 = AD$

| AB \ CD | 00 | 01 | 11 | 10 |
|---|---|---|---|---|
| 00 | $D_0$ | $D_1$ | $D_3$ | $D_2$ |
| 01 | $D_4$ | $D_5$ | $D_7$ | $D_6$ |
| 11 | x | x | x | x |
| 10 | $D_8$ | $D_9$ | x | x |

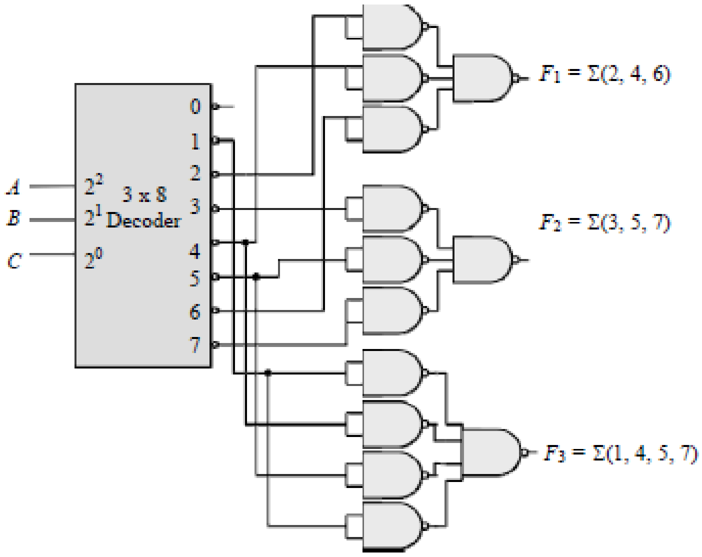
## 4.25



**4.26**

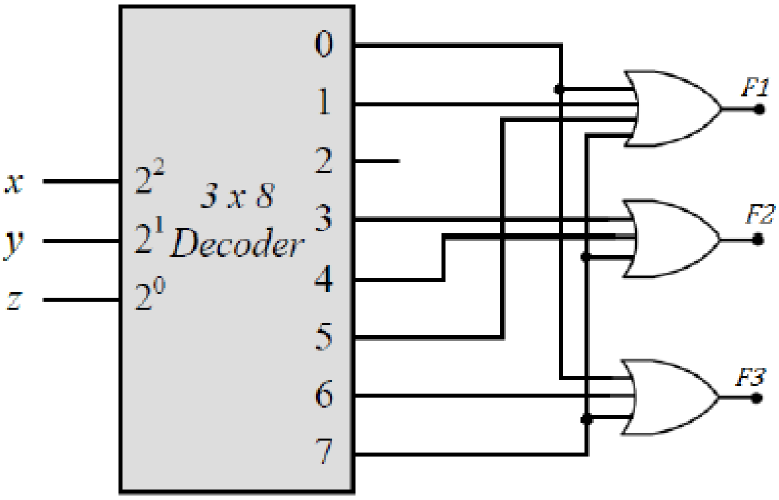


**4.27**




**4.28** **(a)**

$$F_1 = x'y' + xz = x'y'z' + x'y'z + xy'z + xyz = \sum(0,1,5,7)$$
$$F_2 = xy'z' + yz = xy'z' + x'yz + xyz = \Sigma(3,4,7)$$
$$F_3 = x'y'z' + xy = x'y'z' + xyz' + xyz = \Sigma(0,6,7)$$



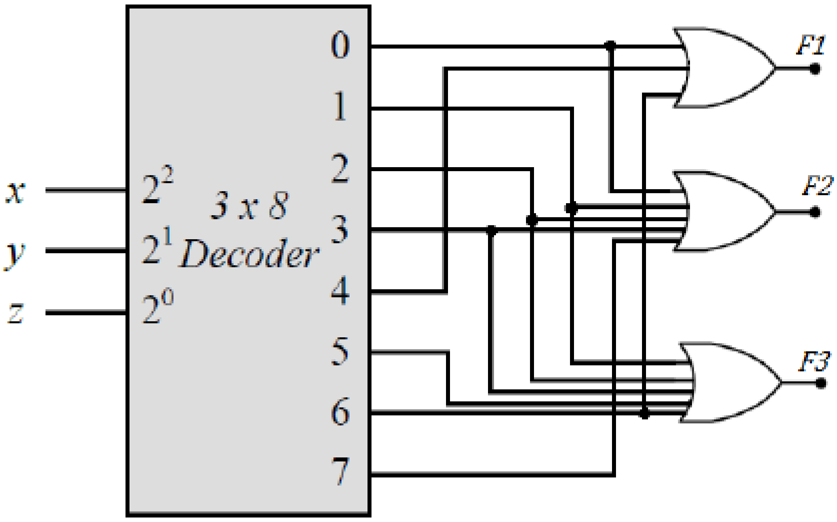
(b) $F_1 = (x + y')z' = xz' + y'z' = xyz' + xy'z' + x'y'z' = \Sigma(0,4,6)$

$F_2 = (x' + y)(x' + z) = x' + x'z + x'y + yz = x' + yz$

$= x'y'z' + x'y'z + x'yz' + x'yz + xyz = \Sigma(0,1,2,3,7)$

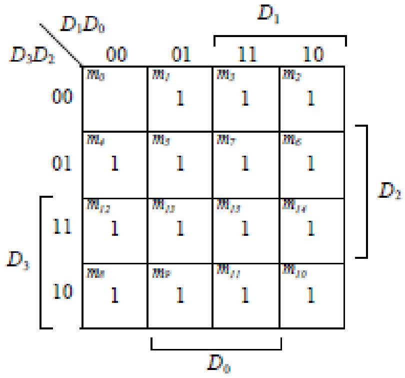
$F_3 = (y + z)(x' + z') = x'y + yz' + x'z$

$= x'yz' + x'yz + xyz' + x'yz' + x'y'z + x'yz = \Sigma(1,2,3,6)$

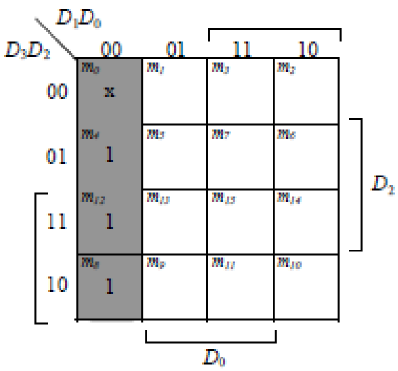


**4.29**

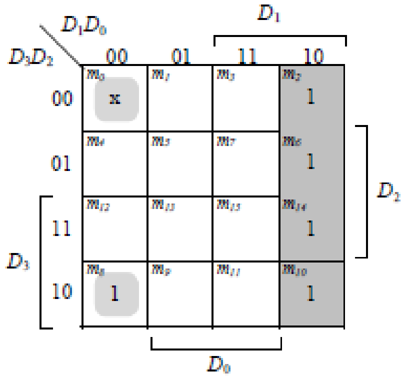
| *Inputs* $D_3$ | $D_2$ | $D_1$ | $D_0$ | *Outputs* $x$ | $y$ | $V$ |
|---|---|---|---|---|---|---|
| 0 | 0 | 0 | 0 | x | x | 0 |
| x | x | x | 1 | 0 | 0 | 1 |
| x | x | 1 | 0 | 0 | 1 | 1 |
| x | 1 | 0 | 0 | 1 | 0 | 1 |
| 1 | 0 | 0 | 0 | 1 | 1 | 1 |



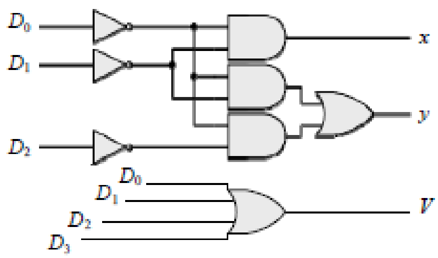

$V = D_0 + D_1 + D_2 + D_3$




$x = D_1'D_0'$




$y = D_0'D_2' + D_1D_0'$

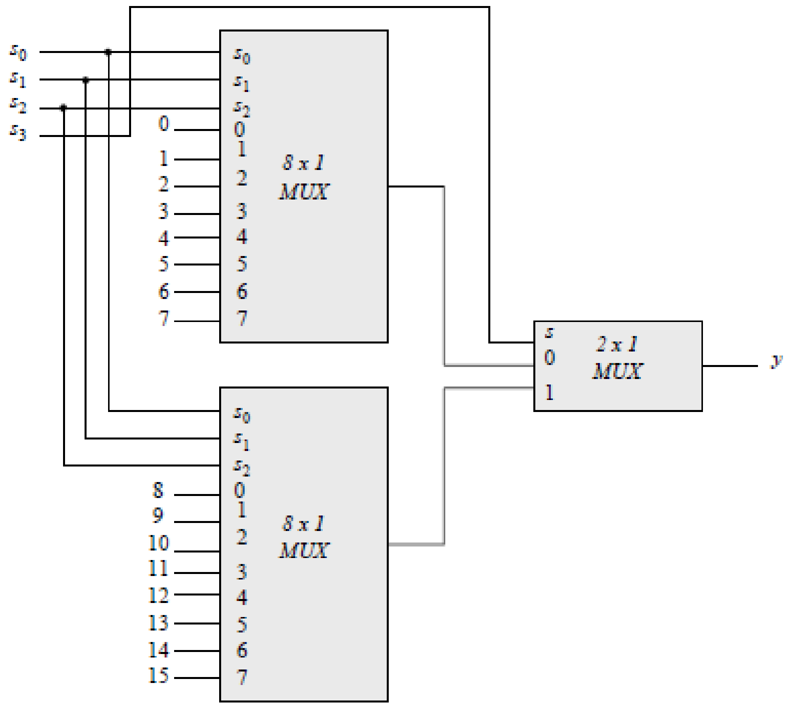


**4.30**

| Inputs | | | | | | | | Outputs | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| $D_0$ | $D_1$ | $D_2$ | $D_3$ | $D_4$ | $D_5$ | $D_6$ | $D_7$ | x | y | z | V |
| 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | x | x | x | 0 |
| 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 1 |
| x | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 1 |
| x | x | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 1 |
| x | x | x | 1 | 0 | 0 | 0 | 0 | 0 | 1 | 1 | 1 |
| x | x | x | x | 1 | 0 | 0 | 0 | 1 | 0 | 0 | 1 |
| x | x | x | x | x | 1 | 0 | 0 | 1 | 0 | 1 | 1 |
| x | x | x | x | x | x | 1 | 0 | 1 | 0 | 0 | 1 |
| x | x | x | x | x | x | x | 1 | 1 | 1 | 1 | 1 |

*If $D_2 = 1$, $D_6 = 1$, all others = 0*
*Output xyz = 100 and V = 1*

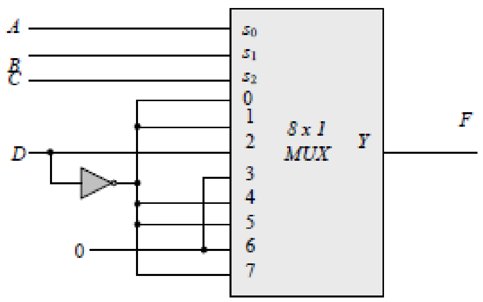
**4.31**



**4.32** (a) $F = \Sigma\,(0, 2, 5, 8, 10, 14)$

$F = \Sigma(0, 2, 5, 8, 10, 14)$

| Inputs ABCD | Mux input line (ABC) | Value | F | |
|---|---|---|---|---|
| 000 0 | 0 | 0 | 1 | $F = D'$ |
| 000 1 | 0 | 1 | 0 | |
| 001 0 | 1 | 2 | 1 | $F = D'$ |
| 001 1 | 1 | 3 | 0 | |
| 010 0 | 2 | 4 | 0 | $F = D$ |
| 010 1 | 2 | 5 | 1 | |
| 011 0 | 3 | 6 | 0 | $F = 0$ |
| 011 1 | 3 | 7 | 0 | |
| 100 0 | 4 | 8 | 1 | $F = D'$ |
| 100 1 | 4 | 9 | 0 | |
| 101 0 | 5 | 10 | 1 | $F = D'$ |
| 101 1 | 5 | 11 | 0 | |
| 110 0 | 6 | 12 | 0 | $F = 0$ |
| 110 1 | 6 | 13 | 0 | |
| 111 0 | 7 | 14 | 1 | $F = D'$ |
| 111 1 | 7 | 15 | 0 | |




**(b)**

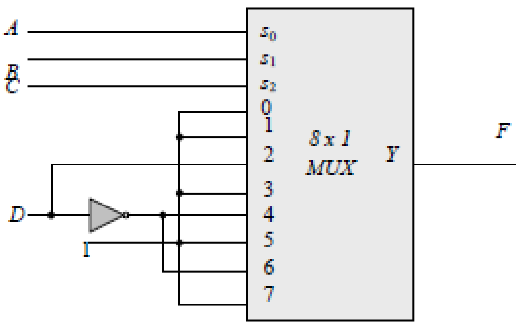
$F = \Pi(2, 6, 11) = (A' + B' + C + D')(A' + B + C + D')(A + B' + C + D)$

$F' = (A' + B' + C + D')' + (A' + B + C + D')' + (A + B' + C + D)'$

$F' = (ABC'D) + (AB'C'D) + (A'BC'D') = \Sigma(13, 9, 4)$

$F = \Sigma(0, 1, 2, 3, 5, 6, 7, 8, 10, 11, 12, 14, 15)$

| Inputs ABCD | Mux input line (ABC) | Value | F | |
|---|---|---|---|---|
| 000 0 | 0 | 0 | 1 | $F = 1$ |
| 000 1 | 0 | 1 | 1 | |
| 001 0 | 1 | 2 | 1 | $F = 1$ |
| 001 1 | 1 | 3 | 1 | |
| 010 0 | 2 | 4 | 0 | $F = D$ |
| 010 1 | 2 | 5 | 1 | |
| 011 0 | 3 | 6 | 1 | $F = 1$ |
| 011 1 | 3 | 7 | 1 | |
| 100 0 | 4 | 8 | 1 | $F = D'$ |
| 100 1 | 4 | 9 | 0 | |
| 101 0 | 5 | 10 | 1 | $F = 1$ |
| 101 1 | 5 | 11 | 1 | |
| 110 0 | 6 | 12 | 1 | $F = D'$ |
| 110 1 | 6 | 13 | 0 | |
| 111 0 | 7 | 14 | 1 | $F = 1$ |
| 111 1 | 7 | 15 | 1 | |



**4.33** Truth Table of a Full Subtractor:

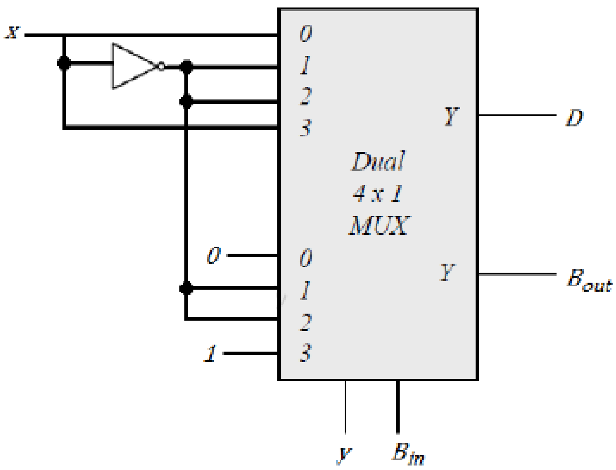
| Inputs | | | Outputs | |
|---|---|---|---|---|
| $x$ | $y$ | $B_{in}$ | $D$ | $B_{out}$ |
| 0 | 0 | 0 | 0 | 0 |
| 0 | 0 | 1 | 1 | 1 |
| 0 | 1 | 0 | 1 | 1 |
| 0 | 1 | 1 | 0 | 1 |
| 1 | 0 | 0 | 1 | 0 |
| 1 | 0 | 1 | 0 | 0 |
| 1 | 1 | 0 | 0 | 0 |
| 1 | 1 | 1 | 1 | 1 |

$x$: Minuend, $y$: subtrahend, $B_{in}$: Borrow In, $D$: Difference, $B_{out}$: Borrow Out

$D(x, y, B_{in}) = \Sigma(1,2,4,7)$
$B_{out}(x, y, B_{in}) = \Sigma(1,2,3,7)$

| $D$ | $I_0$ | $I_1$ | $I_2$ | $I_3$ |
|---|---|---|---|---|
| $x'$ | 0 | 1 | 2 | 3 |
| $x$ | 4 | 5 | 6 | 7 |
| | $x$ | $x'$ | $x'$ | $x$ |

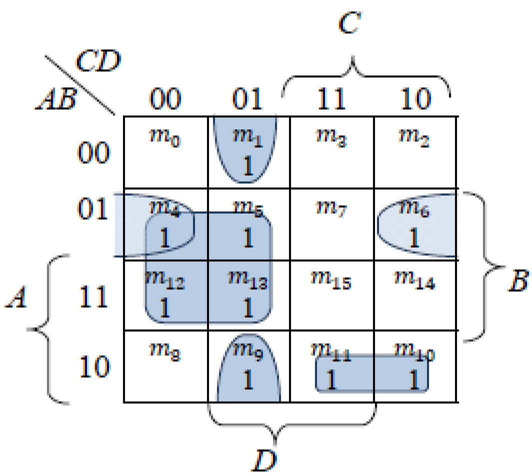
| $B_{out}$ | $I_0$ | $I_1$ | $I_2$ | $I_3$ |
|---|---|---|---|---|
| $x'$ | 0 | 1 | 2 | 3 |
| $x$ | 4 | 5 | 6 | 7 |
| | 0 | $x'$ | $x'$ | 1 |



**4.34** (a)

| $A$ | $B$ | $C$ | $D$ | $I$ | $F$ |
|---|---|---|---|---|---|
| 0 | 0 | 0 | 0 | $I_0 = D$ | 0 |
| 0 | 0 | 0 | 1 | | 1 |
| 0 | 0 | 1 | 0 | $I_1 = 0$ | 0 |
| 0 | 0 | 1 | 1 | | 0 |
| 0 | 1 | 0 | 0 | $I_2 = 1$ | 1 |
| 0 | 1 | 0 | 1 | | 1 |
| 0 | 1 | 1 | 0 | $I_3 = D'$ | 1 |
| 0 | 1 | 1 | 1 | | 0 |
| 1 | 0 | 0 | 0 | $I_4 = D$ | 0 |
| 1 | 0 | 0 | 1 | | 1 |
| 1 | 0 | 1 | 0 | $I_5 = 1$ | 1 |
| 1 | 0 | 1 | 1 | | 1 |
| 1 | 1 | 0 | 0 | $I_6 = 1$ | 1 |
| 1 | 1 | 0 | 1 | | 1 |
| 1 | 1 | 1 | 0 | $I_7 = 0$ | 0 |
| 1 | 1 | 1 | 1 | | 0 |

$F = \Sigma(1,4,5,6,9,10,11,12,13)$

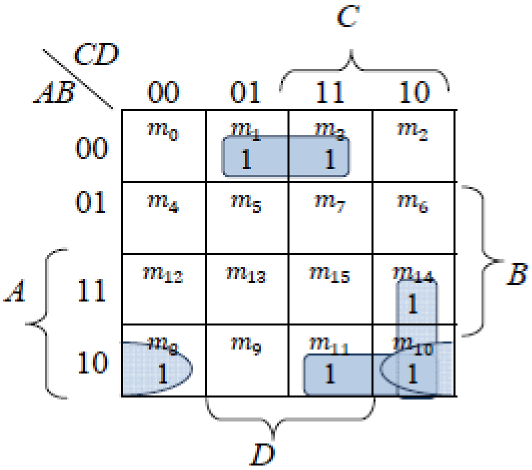



$F = A'BD' + AB'C + B'C'D + BC'$

**(b)**

| $A$ | $B$ | $C$ | $D$ | $I$ | $F$ |
|---|---|---|---|---|---|
| 0 | 0 | 0 | 0 | $I_0 = D$ | 0 |
| 0 | 0 | 0 | 1 | | 1 |
| 0 | 0 | 1 | 0 | $I_1 = D$ | 0 |
| 0 | 0 | 1 | 1 | | 1 |
| 0 | 1 | 0 | 0 | $I_2 = 0$ | 0 |
| 0 | 1 | 0 | 1 | | 0 |
| 0 | 1 | 1 | 0 | $I_3 = 0$ | 0 |
| 0 | 1 | 1 | 1 | | 0 |
| 1 | 0 | 0 | 0 | $I_4 = D'$ | 1 |
| 1 | 0 | 0 | 1 | | 0 |
| 1 | 0 | 1 | 0 | $I_5 = 1$ | 1 |
| 1 | 0 | 1 | 1 | | 1 |
| 1 | 1 | 0 | 0 | $I_6 = 0$ | 0 |
| 1 | 1 | 0 | 1 | | 0 |
| 1 | 1 | 1 | 0 | $I_7 = D'$ | 1 |
| 1 | 1 | 1 | 1 | | 0 |

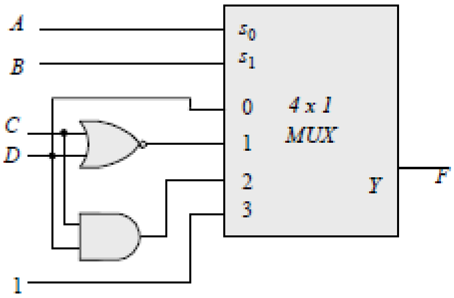
$F = \Sigma(1,3,8,10,11,14)$




$F = A'B'D + AB'D' + AB'C + ACD'$

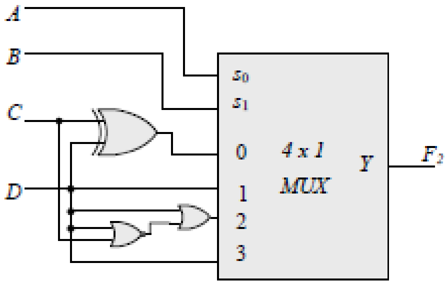
**4.35** **(a)**

| Inputs ABCD | F | |
|---|---|---|
| 0000 | 0 | |
| 0001 | 1 | $AB = 00$ |
| 0010 | 0 | $F = D$ |
| 0011 | 1 | |
| 0100 | 1 | $AB = 01$ |
| 0101 | 0 | $F = C'D'$ |
| 0110 | 0 | $= (C + D)'$ |
| 0111 | 0 | |
| 1000 | 0 | |
| 1001 | 0 | $AB = 10$ |
| 1010 | 0 | $F = CD$ |
| 1011 | 1 | |
| 1100 | 1 | $AB = 11$ |
| 1101 | 1 | $F = 1$ |
| 1110 | 1 | |
| 1111 | 1 | |

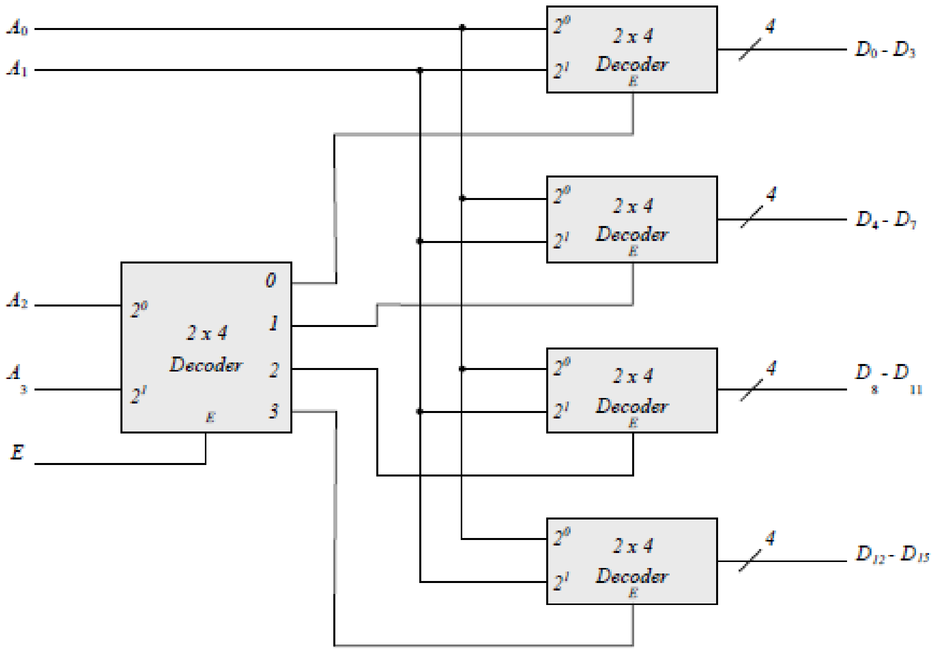



**(b)** $F = \Sigma(1, 2, 5, 7, 8, 10, 11, 13, 15)$

| Inputs ABCD | $F_2 = \Sigma(1, 2, 5, 7, 8, 10, 11, 13, 15)$ | |
|---|---|---|
| 0000 | 0 | |
| 0001 | 1 | $AB = 00$ |
| 0010 | 1 | $F = C'D + CD'$ |
| 0011 | 0 | |
| 0100 | 0 | $AB = 01$ |
| 0101 | 1 | $F = C'D + CD = D$ |
| 0110 | 0 | |
| 0111 | 1 | |
| 1000 | 1 | |
| 1001 | 0 | |
| 1010 | 1 | $AB = 10$ |
| 1011 | 1 | $F = C'D' + C'D + CD = C'D' + D$ |
| 1100 | 0 | $AB = 11$ |
| 1101 | 1 | $F = D$ |
| 1110 | 0 | |
| 1111 | 1 | |



**4.63**

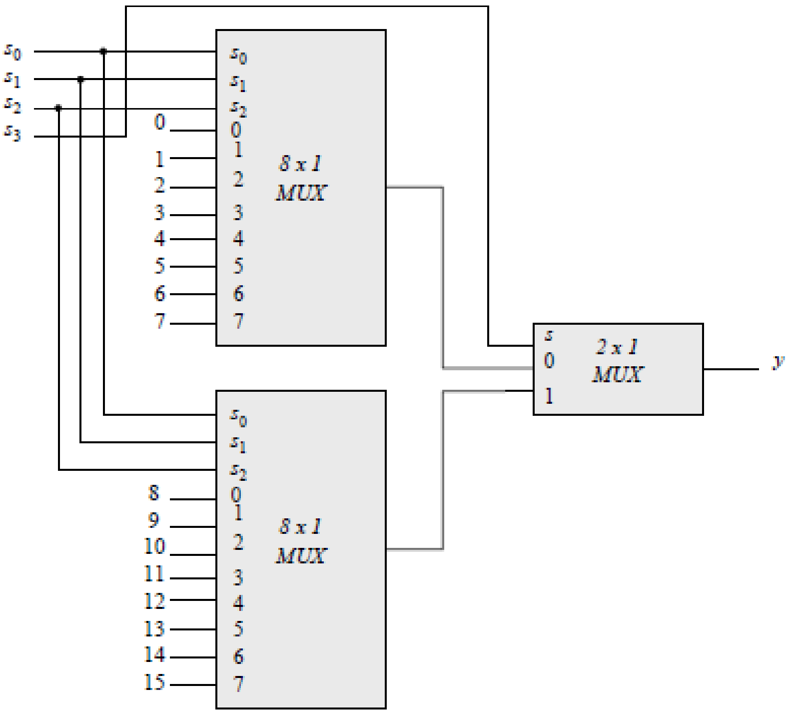



**4.64**

| Inputs | | | | | | | | Outputs | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| $D_0$ | $D_1$ | $D_2$ | $D_3$ | $D_4$ | $D_5$ | $D_6$ | $D_7$ | $x$ | $y$ | $z$ | $V$ |
| 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | x | x | x | 0 |
| 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 1 |
| x | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 1 |
| x | x | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 1 |
| x | x | x | 1 | 0 | 0 | 0 | 0 | 0 | 1 | 1 | 1 |
| x | x | x | x | 1 | 0 | 0 | 0 | 1 | 0 | 0 | 1 |
| x | x | x | x | x | 1 | 0 | 0 | 1 | 0 | 1 | 1 |
| x | x | x | x | x | x | 1 | 0 | 1 | 0 | 0 | 1 |
| x | x | x | x | x | x | x | 1 | 1 | 1 | 1 | 1 |

*If $D_2 = 1$, $D_6 = 1$, all others = 0*
*Output $xyz = 100$ and $V = 1$*

**4.65**



$s_0$
$s_1$
$s_2$
$s_3$
0 1 2 3 4 5 6 7
8 x 1 MUX
8 9 10 11 12 13 14 15
8 x 1 MUX
2 x 1 MUX
s
0
1
y